

Assignment 3 Part 1 — Team Task Breakdown

CPRG307 | Aaron Reid, Amal Musse, Kaley Wood | February 3, 2026

Assignment Overview

Write a single PL/SQL anonymous block for the WKIS accounting system that reads pending transactions from NEW_TRANSACTIONS, inserts them into TRANSACTION_HISTORY and TRANSACTION_DETAIL, updates ACCOUNT balances using double-entry accounting rules, and deletes each processed row from the holding table.

Loop Context Legend

DECLARE	Task belongs to the DECLARE section — cursor definitions and variable declarations
Outer Loop	Task runs once per transaction — inside the outer cursor loop
Inner Loop	Task runs once per detail line — inside the inner cursor loop (nested)
Both Loops	Task spans both loops — applies during testing / full execution

Member Roles

Member	Task & Sub-tasks	Loop Context
Aaron Reid	- Program header documentation - Authors, date, course, purpose block comment	DECLARE
	- Outer explicit cursor definition - SELECT DISTINCT transaction_number from NEW_TRANSACTIONS - Declared in the DECLARE section	DECLARE
	- Outer FOR loop structure - One iteration = one full transaction - Drives the inner loop and all DML for that transaction	Outer Loop
	- INSERT into TRANSACTION_HISTORY - One row per transaction (not per detail line) - Uses: transaction_number, transaction_date, description from outer cursor - COMMIT placement - Placed after all DML for one transaction completes - Ensures each transaction succeeds or fails as a unit	Outer Loop
Amal Musse	- Inner explicit cursor definition - SELECT all detail rows WHERE transaction_number = outer cursor value - Declared in the DECLARE section, dependent on outer cursor	DECLARE
	- Inner FOR loop structure - One iteration = one detail line of the current transaction - Nested inside the outer loop	Inner Loop
	- INSERT into TRANSACTION_DETAIL - One row per detail line	Inner Loop

Member	Task & Sub-tasks	Loop Context
	◦ Uses: account_number, amount, transaction_type from inner cursor • DELETE from NEW_TRANSACTIONS ◦ Removes each processed row using inner cursor data ◦ Executed inside the inner loop, once per detail line	Inner Loop
Kaley Wood	• SELECT INTO — account default type lookup ◦ Queries ACCOUNT table for the default transaction type (D or C) ◦ Performed inside the inner loop using the current account number	Inner Loop
	• UPDATE ACCOUNT balances ◦ Add amount if transaction type matches account default type ◦ Subtract amount if transaction type does not match account default type	Inner Loop
	• Create additional test data ◦ Covers debit account credited, credit account debited ◦ Multi-line transactions, single-line transactions	Both Loops
	• Integration testing ◦ Verify all 4 tables update correctly after each run ◦ Refresh data between executions ◦ Confirm NEW_TRANSACTIONS rows are fully removed	Both Loops

Shared by All Members

- Agree on cursor column names and variable naming before writing code
- Add inline comments throughout the merged script
- Final review — one anonymous block, no stored programs, no hard-coded values, no GOTO/EXIT/SAVEPOINT
- Complete peer evaluations on Brightspace by the due date